

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Miniature Mass Spectrometers and Front-End Interfaces

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ABSTRACT

Mass spectrometers (MSs) are workhorse instruments on practically every robotic mission to planetary objects in our solar system. As long-duration human flight to other planets is planned, one also needs to monitor a cabin atmosphere over a period of several years against the toxic buildup of gases. Miniaturized MSs and front ends (gas chromatographs, liquid interfaces, drills, etc.) are needed in these harsh environments. Robustness and reliability are required, and mass, power, and volume are at a premium (especially for missions to the outer planets, including Pluto). We will present work carried out in the JPL Atomic and Molecular Collisions Group and at Thorleaf Research on (1) miniaturization of a quadrupole mass-spectrometer array (QMSA), which has been aboard the International Space Station since 2/01 to monitor possible leaks of cabin air, ammonia, and hydrazines in low-earth orbit; (2) a miniature hyperboloidal Paul ion trap; (3) a miniature front-end gas chromatograph and preconcentrator for use in complex gas mixtures, and modular to the mass spectrometers in (1) and (2). Properties of the systems will be discussed, such as resolution (0.03-0.8 amu, FWHM), sensitivity (10^{12} to 10^{14} counts/torr-s) flow rates, and pumping. The limits to resolution and sensitivity, and the utility of building even smaller MS systems, will be discussed. This work was carried out at the Jet Propulsion Laboratory/California Institute of Technology, and was supported under contract with the National Aeronautics and Space Administration.